SOCIOL 723: Social Statistics II

Week 10, Panel Data and Difference-in-Differences

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Today

Panel Data and Unobserved Heterogeneity

Difference-in-Differences

The Trouble with Two-Way Fixed Effects

★ = essential, you must understand this ★ = for understanding, not examined

A Third Source of Variation

So far: adjust for observables (Week 6), find an instrument (Week 7), find a threshold (Week 9). Today: use variation over time within the same unit.

The core idea

If we observe the same unit before and after, we can difference away everything about that unit that does not change, including confounders we never measured and could not name.

- That is a genuinely powerful trick, and it is why panel data are prized.
- It buys nothing against confounders that *do* change over time, and the price is a new and demanding assumption about counterfactual trends.
- The last decade has shown that the standard implementation, two-way fixed effects, is badly behaved in exactly the setting sociologists use most.

Panel Data

The Error Components Model ★

With unit i observed at times $t = 1, \ldots, T$:

$$Y_{it} = \beta D_{it} + \underbrace{\alpha_i}_{\text{unit effect}} + \underbrace{\lambda_t}_{\text{time effect}} + \epsilon_{it}.$$

- α_i absorbs *every* time-invariant characteristic of unit i , observed or not, nameable or not.
- Pooled OLS is biased whenever $\text{Cov}(D_{it}, \alpha_i) \neq 0$: states that adopt a policy differ persistently from states that do not.
- **Random effects** assumes $\text{Cov}(D_{it}, \alpha_i) = 0$ and is more efficient if true. It is almost never true in the settings we care about. The Hausman test compares the two; when in doubt, use fixed effects.

The Within Transformation, Explicitly ★

Average the model within unit and subtract:

$$
\begin{aligned}
Y_{it} &= \beta D_{it} + \alpha_i + \lambda_t + \epsilon_{it} \\
\bar{Y}_{i\cdot} &= \beta \bar{D}_{i\cdot} + \alpha_i + \bar{\lambda} + \bar{\epsilon}_{i\cdot} \\
\Longrightarrow \quad Y_{it} - \bar{Y}_{i\cdot} &= \beta(D_{it} - \bar{D}_{i\cdot}) + (\lambda_t - \bar{\lambda}) + (\epsilon_{it} - \bar{\epsilon}_{i\cdot}).
\end{aligned}
$$

- α_i is gone, whatever it was. That is the whole appeal: we never had to name it, measure it, or model it.
- Anything time-invariant is annihilated with it. You cannot estimate the effect of race, sex, or birth cohort in a unit-fixed-effects model, not because they do not matter but because there is no within-unit variation.
- Two-way fixed effects demeans within unit *and* within period, removing common shocks.

The Within Transformation, Explicitly ★ (cont.)

- The equivalence with unit dummies is exactly FWL from Week 1: regressing on unit dummies partials out the unit means.

Random Effects, and Why We Usually Do Not Use It

Random effects treats α_i as a draw from a distribution and assumes the unit effect is unrelated to the regressors. The slogan version is $\mathrm{Cov}(D_{it}, \alpha_i) = 0$, but consistency needs the stronger statement

$$E\left[\alpha_i \mid D_{i1}, \ldots, D_{iT}\right] = 0,$$

mean independence from the *whole history* of the regressor, together with strict exogeneity of the idiosyncratic error, $E[\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i] = 0$. It then estimates by GLS, a matrix-weighted average of the within and between estimators.

- **If the assumption holds**, RE is more efficient and can identify coefficients on time-invariant covariates. That is a real gain.
- **The assumption says states that adopt a policy are, on average, no different from states that do not.** In the settings sociologists study it is almost never plausible.

Random Effects, and Why We Usually Do Not Use It (cont.)

- The **Hausman test** compares the two: a large difference indicates $\mathrm{Cov}(D, \alpha) \neq 0$ and favours FE. But failing to reject is weak evidence, as usual.
- The default should be fixed effects, with random effects reserved for cases where you can defend the exogeneity of the unit effect, or where the multilevel structure is itself the object of interest.

Nickell Bias and the FE-versus-LDV Choice

Sociologists routinely control for last year's outcome, to absorb differences between units that are *changing* rather than fixed. It is a reasonable instinct. But combining that move with unit fixed effects creates a new bias, and it is worth knowing why before you reach for it.

Add a lagged dependent variable to a fixed-effects model:

$$Y_{it} = \rho Y_{i,t-1} + \beta D_{it} + \alpha_i + \epsilon_{it}.$$

Demeaning makes $Y_{i,t-1} - \bar{Y}_{i\cdot}$ mechanically correlated with $\epsilon_{it} - \bar{\epsilon}_{i\cdot}$, because $\bar{Y}_{i\cdot}$ contains Y_{it}, which contains ϵ_{it}.

- The resulting Nickell bias is $O(1/T)$: severe in short panels ($T = 3$), negligible in long ones. Arellano–Bond-type GMM instruments the lagged difference with deeper lags.

Nickell Bias and the FE-versus-LDV Choice (cont.)

- **FE and LDV embody different selection stories**: FE assumes selection on time-invariant traits; LDV assumes selection on prior outcomes. They can both be valid, but they are different assumptions and will generally give different answers.
- Angrist and Pischke (2009) suggest running both: under some conditions the two bracket the truth, so agreement is reassuring and disagreement tells you how much the selection story matters. Treat that as a heuristic, not a theorem.

Three Ways to Remove α_i

Estimator	Transformation	Note
Within (fixed effects)	$Y_{it} - \bar{Y}_i = \beta(D_{it} - \bar{D}_i) + \tilde{\epsilon}_{it}$	the default
First difference	$\Delta Y_{it} = \beta \Delta D_{it} + \Delta \epsilon_{it}$	equals within when $T = 2$
Unit dummies	include $n - 1$ indicators	numerically identical to within

- All three remove α_i; the first two are transformations, the third does it with parameters. The costs (previous frames) are shared.
- Within and first differences differ when $T > 2$ and disagree under serial correlation; report which you used and why.

What Fixed Effects Costs ★

- **Precision**. You are using only within-unit variation. If D_{it} barely moves within units, the standard error explodes, FWL again.
- **Measurement error is amplified**. Differencing removes signal and keeps noise, so attenuation bias gets worse.
- **Nickell bias and the FE-versus-LDV choice**, as on the previous frames: adding a lagged outcome to a fixed-effects model creates its own bias, and FE and LDV embody different selection stories.
- Always cluster on the unit (Week 2). Serial correlation within units is the norm.

DID

The Canonical 2×2 ★

Two groups, two periods. One group is treated in period 2.

	Before	After	Difference
Treated	$E[Y \mid g{=}1, t{=}1]$	$E[Y \mid g{=}1, t{=}2]$	Δ_1
Control	$E[Y \mid g{=}0, t{=}1]$	$E[Y \mid g{=}0, t{=}2]$	Δ_0
			$\tau = \Delta_1 - \Delta_0$

- The first difference removes anything fixed about each group.
- The second difference removes anything common to both groups over time.
- Equivalent regression: $Y_{it} = \alpha + \gamma\,\text{treated}_i + \delta\,\text{post}_t + \tau\,(\text{treated}_i \times \text{post}_t) + \epsilon_{it}$.

The 2×2 in Regression Form

Write $g_i \in \{0, 1\}$ for group and $t \in \{1, 2\}$ for period:

$$Y_{it} = \alpha + \gamma\, g_i + \delta\, \mathrm{post}_t + \tau\, (g_i \times \mathrm{post}_t) + \epsilon_{it}.$$

Read off the four cell means:

	Before	After	Difference
Treated ($g = 1$)	$\alpha + \gamma$	$\alpha + \gamma + \delta + \tau$	$\delta + \tau$
Control ($g = 0$)	α	$\alpha + \delta$	δ
			τ

- γ absorbs fixed differences in level; δ absorbs the common time trend; τ is what is left.

The 2×2 in Regression Form (cont.)

- With unit and period fixed effects instead of g_i and post_t; this is the two-way fixed-effects model, identical when there are two groups and two periods, and, as we will see, not identical otherwise.

Parallel Trends ★

The identifying assumption

$$E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 0].$$

Absent treatment, the two groups' average outcomes would have moved in parallel.

- Levels may differ arbitrarily. Only the *trends* in the untreated potential outcome must agree.
- Untestable: it concerns a counterfactual trend. Pre-treatment parallelism is suggestive evidence, not proof.
- **Not scale-free**: parallel trends in levels and parallel trends in logs are different assumptions, and both cannot generally hold. The next slide shows why this is a real choice and not a technicality.

Parallel Trends ★ (cont.)

- Needs **no anticipation**: units must not change behaviour before treatment in expectation of it.

What the Double Difference Identifies, Exactly ★

Decompose the treated group's observed change. Its period-2 outcome is $Y_{i2}(1)$; add and subtract the counterfactual $Y_{i2}(0)$:

$$\Delta_1 = E[Y_{i2}(1) - Y_{i1}(0) \mid g_i{=}1] = \underbrace{E[Y_{i2}(1) - Y_{i2}(0) \mid g_i{=}1]}_{\text{ATT}} + \underbrace{E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}1]}_{\text{treated group's } \textit{untreated} \text{ trend}}.$$

The control group supplies an estimate of that second, unobservable term: $\Delta_0 = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}0]$. Subtracting,

$$\boxed{\Delta_1 - \Delta_0 \;=\; \text{ATT} \;+\; \underbrace{E[\Delta Y(0) \mid g{=}1] - E[\Delta Y(0) \mid g{=}0]}_{\text{differential trend; } =0 \text{ under parallel trends}}}$$

- The estimand is the ATT, not the ATE: we borrowed a counterfactual trend *for the treated group*. Nothing in the design identifies what treatment would have done to the controls.

What the Double Difference Identifies, Exactly ★ (cont.)

- The bias term *is* the parallel-trends violation, exactly. Every diagnostic in this lecture (pre-trends, placebos, Rambachan–Roth bounds) is aimed at that one object.
- Note also what was *not* assumed: no restriction on the level difference γ, and no randomization anywhere.

Parallel Trends Is Not Scale-Free ★

Suppose untreated outcomes would have grown by 10% in both groups, from a base of 20 in the treated group and 40 in the control group.

	Before	After	Change
Treated	20	22	+2
Control	40	44	+4

- Parallel trends holds in logs (both +10%) but fails in levels (+2 versus +4). A DID in levels would report a spurious -2.
- The two assumptions are mutually exclusive unless base levels are equal. There is no neutral choice, and the transformation is part of the identifying assumption, not a presentational detail.

Parallel Trends Is Not Scale-Free ★ (cont.)

- Choose on substantive grounds, is the intervention plausibly additive or multiplicative?, and *before* looking at the results.
- One diagnostic: if pre-treatment levels are far apart, be especially careful. If they are similar, the choice matters less.

Event Studies ★

Generalize to many periods by estimating an effect for each period relative to treatment:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbf{1}\{t - g_i = k\} + \epsilon_{it},$$

where g_i is unit i's treatment date and $k = -1$ is the omitted reference.

- **Pre-period coefficients** ($k < 0$) are the parallel-trends check. Flat and near zero is what you want to see.
- **Post-period coefficients** trace out the dynamics: does the effect grow, fade, or appear only after a lag?
- Caution: a flat pre-trend is weak evidence. The test often has low power, and conditioning your analysis on passing it distorts inference (Roth 2022). Report the power of your pre-trend test.

Event Studies: Specification Details That Matter

That specification looks innocent. Three choices inside it change the picture, and each must be reported.

- **Normalization**: one lead must be omitted (here $k = -1$). Every τ_k is relative to it, so the picture shifts if you change the reference period. Say which you used.
- **Binning the endpoints**: with unbalanced exposure, far leads and lags are estimated off few units. Bin them ($k \leq -5$, $k \geq 5$) or restrict to a balanced window, and say which.
- **Never-treated units** anchor the time effects λ_t. If there are none, the λ_t are identified only off variation in timing, and the contamination problem below is unavoidable.

Staggered

The Setting Everyone Uses ★

Units adopt the policy at *different times*: states pass a law in 1995, 2001, 2008. The standard specification is

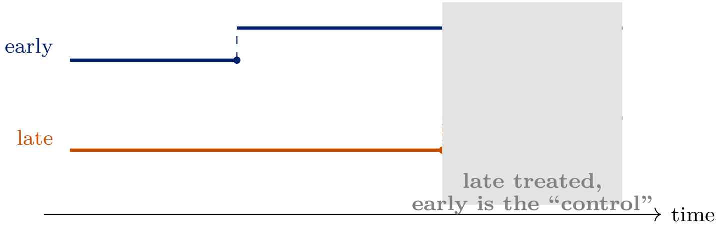
$$Y_{it} = \alpha_i + \lambda_t + \beta\, D_{it} + \epsilon_{it}, \qquad D_{it} = \mathbf{1}\{t \geq g_i\}.$$

For thirty years this was assumed to estimate "the" treatment effect. It does when the effect is constant across units and over time since treatment , but with heterogeneous or dynamic effects it need not estimate any sensible average at all.

Goodman-Bacon (2021)

The TWFE estimate is a weighted average of *all possible* 2×2 DID comparisons in the data, including comparisons that use already-treated units as controls.

The Forbidden Comparison ★



- In the shaded window, TWFE uses the *early-treated* unit as a control for the late-treated one.
- If the early unit's effect is still evolving, that evolution is subtracted from the late unit's effect, and enters with a negative weight.
- With dynamic effects, TWFE can be the wrong *sign* even when every unit-level effect is positive.

The Forbidden Comparison, in Algebra ★

Take the late-versus-early 2×2 from the shaded window: late switches from untreated to treated, early is *treated throughout*. Suppose parallel trends holds exactly and untreated outcomes are flat, so every change in Y is a treatment effect. Then

$$\hat{\beta}^{2\times 2}_{\text{late vs. early}} = \underbrace{\text{ATT}_{\text{late}}(\text{post})}_{\text{what we want}} - \underbrace{\Big[\text{ATT}_{\text{early}}(\text{post}) - \text{ATT}_{\text{early}}(\text{pre})\Big]}_{\text{the } \textit{growth} \text{ in early's effect over the window}} .$$

With numbers; no noise, no trends, every true effect positive:

	$t=3$	$t=4$	Change
Late (treated at t=4; effect 1)	0	1	+1
Early (treated at t=2; effects $1,2,3$)	2	3	+1
			$\hat{\beta}^{2\times 2} = 0$

The Forbidden Comparison, in Algebra ★ (cont.)

- Late's true effect is $+1$; this comparison returns 0. If early's effect grew by 2 per period instead, it would return -1.
- If effects were constant $(1, 1, 1)$, the bracketed term would vanish and the comparison would be harmless, which is exactly why the problem stayed invisible for thirty years of constant-effect thinking.

The Goodman-Bacon Decomposition ★

With staggered timing, the TWFE estimate is a weighted average of every possible 2×2 comparison in the data:

$$\hat{\beta}^{\text{TWFE}} = \sum_{\text{pairs}} s_{ab}\, \hat{\beta}_{ab}^{2\times 2}, \qquad \sum s_{ab} = 1,$$

where the weights s_{ab} depend on group sizes and on the *variance of treatment* within each pair, largest for pairs whose treatment switches mid-window. (The next frame writes them out.)

Three kinds of comparison arise:

1. Treated versus **never-treated**, clean.
2. **Earlier** treated versus **later** treated, before the later group switches, clean.
3. **Later** treated versus **earlier** treated, after the earlier group switches, the forbidden comparison.

Type 3 uses already-treated units as controls. If their effects are still evolving, that evolution is subtracted from the late group's effect.

The Goodman-Bacon Weights, Explicitly ★

Let n_k, n_l, n_U be the groups' sample shares and $\bar{D}_j$ the fraction of the panel that group j spends treated (k treated earlier than l). Up to a common normalization, the weight on each 2×2 is the product of the pair's group sizes and the *variance of the treatment dummy within that pair's window*:

$$s_{kU} \propto n_k\, n_U\, \bar{D}_k(1 - \bar{D}_k) \qquad \text{treated vs. never-treated}$$

$$s_{kl}^k \propto n_k\, n_l\, (\bar{D}_k - \bar{D}_l)(1 - \bar{D}_k) \qquad \text{early vs. late, before late switches}$$

$$s_{kl}^l \propto n_k\, n_l\, \bar{D}_l(\bar{D}_k - \bar{D}_l) \qquad \text{late vs. early, after early switches}$$

- Every weight has the $\bar{D}(1 - \bar{D})$ shape of a Bernoulli variance, so groups treated near the middle of the panel dominate: the weights come from OLS mechanics, not from group sizes alone and not from any estimand you chose. Compare Week 1's variance-weighting result: same phenomenon, new costume.

The Goodman-Bacon Weights, Explicitly ★ (cont.)

- The forbidden weight s_{kl}^{l} is not negligible in general: it grows with $\bar{D}_k$, i.e. with how long the early group has already been treated when the late group switches.
- All of this is computable from the data alone: `bacondecomp::bacon()` lists every 2×2, its estimate, and its weight. We decompose a real TWFE estimate in lab.

Where the Negative Weights Come From ★

FWL (Week 1) reduces TWFE to a bivariate regression on the *double-demeaned* treatment. With $\bar{D}_i$, $\bar{D}_t$, $\bar{D}$ the unit, period, and grand means of D_{it}:

$$\hat{\beta}^{\text{TWFE}} = \frac{\sum_{it} \tilde{D}_{it} Y_{it}}{\sum_{it} \tilde{D}_{it}^2}, \qquad \tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}.$$

Substitute $Y_{it} = \alpha_i + \lambda_t + \tau_{it} D_{it} + \epsilon_{it}$: the fixed effects are annihilated because $\tilde{D}$ has zero unit and period sums, and $\sum_{it} \tilde{D}_{it}^2 = \sum_{(i,t):D_{it}=1} \tilde{D}_{it}$, so

$$\text{plim}\ \hat{\beta}^{\text{TWFE}} = \sum_{(i,t):\, D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{(i,t):\, D_{it}=1} \tilde{D}_{it}}, \qquad \sum w_{it} = 1.$$

- For a *treated* cell, $\tilde{D}_{it} = 1 - \bar{D}_i - \bar{D}_t + \bar{D}$, which is negative exactly when $\bar{D}_i + \bar{D}_t > 1 + \bar{D}$: a unit treated for most of the panel, observed in a period when most of the sample is treated. That is the early adopter, late in the panel.

Where the Negative Weights Come From ★ (cont.)

- Intuition: OLS rewards cells whose treatment status is *surprising* given their unit and period averages. An early adopter's late-period treatment is completely unsurprising, so its effect is priced at zero, or below zero.
- The weights are a mechanical fact about OLS and are there even under constant effects, harmlessly. Heterogeneous τ_{it} is what turns strange weights into a wrong answer.

Negative Weights: Consequences and Diagnostics ★

de Chaisemartin and D'Haultfœuille (2020) generalize the previous frame's calculation, allowing covariates and non-binary treatments, and show that negative weights are the rule in staggered designs, not a corner case.

- Consequence: TWFE can have the wrong sign even when every $\tau_{it} > 0$. This is not a small-sample problem; it survives $n \to \infty$, because the weights, not the noise, are at fault.
- By the formula, the negative weights fall on units treated *early* and observed over a *long* post-period, exactly the units carrying the most information about long-run effects. TWFE quietly discounts, and then inverts, the long run.

Negative Weights: Consequences and Diagnostics ★ (cont.)

- **Diagnostic**: compute the weights (`TwoWayFEWeights::twowayfeweights`) and report how many are negative and what share of the total mass they carry. Small negative mass is reassuring about *this particular* problem; it says nothing about parallel trends or about dynamic-effect contamination, so it is one diagnostic, not a validity test.
- Sun and Abraham (2021) show the same contamination in event-study coefficients: with staggered timing, the TWFE $\hat{\tau}_k$ at lag k is a weighted combination of *every* cohort's effects at *every* lag, with weights that need not be positive or sum to one lag by lag. So apparent pre-trends can be artefacts of post-treatment dynamics, and a clean pre-trend plot from raw TWFE proves nothing.

TWFE is fine when treatment timing is common, or when effects are genuinely constant. It is unreliable exactly where sociologists use it most.

Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate

Define the group-time average treatment effect for cohort g at time t:

$$ATT(g,t) = E\big[Y_t(g) - Y_t(\infty) \mid G_i = g\big],$$

identified by comparing cohort g to units *not yet treated* at t (or to never-treated units), using only periods before g as the baseline. Under cohort-level parallel trends and no anticipation,

$$ATT(g,t) = E\big[Y_t - Y_{g-1} \mid G_i = g\big] - E\big[Y_t - Y_{g-1} \mid \text{not yet treated at } t\big] :$$

a clean 2×2. The baseline $g-1$ predates cohort g's treatment, the comparison group has not started treatment by t, and no already-treated unit is ever used as a control.

- Under *conditional* parallel trends, each block can be estimated by IPW, outcome regression, or a doubly robust estimator, Week 6, reappearing inside a DID.

Callaway–Sant'Anna: Build Clean Blocks, Then Aggregate (cont.)

- **Then choose an aggregation**, and the choice is an estimand choice:
  - *simple*: average over all treated (g, t);
  - *dynamic*: average by $t - g$, giving an event-study path;
  - *group*: average within cohort, showing who benefited;
  - *calendar*: average within period, showing when.
- Sun–Abraham reaches the same place through an interaction-weighted regression; Borusyak–Jaravel–Spiess by imputing $Y(0)$ from untreated cells.

The Modern Estimators ★

All of them share one principle: never use an already-treated unit as a control. Build clean 2×2 comparisons, then aggregate with weights you choose.

Method	Idea	R
Callaway–Sant'Anna (2021)	group-time ATT(g, t), never- or not-yet-treated controls	`did`
Sun–Abraham (2021)	interaction-weighted event study	`fixest::su`
de Chaisemartin–D'Haultfœuille	clean comparisons at switch dates	`DIDmultipl`
Borusyak–Jaravel–Spiess	impute $Y(0)$ from untreated cells	`didimputat`
Stacked / cohort DID	one clean event window per cohort	manual

Callaway–Sant'Anna also allows **conditional** parallel trends: trends parallel only after adjusting for covariates, estimated by IPW, regression, or doubly robust, Week 6, reappearing.

Honest Inference About Pre-Trends

Rather than testing pre-trends and proceeding as if they had passed, Rambachan and Roth (2023) propose bounding how badly parallel trends could fail.

- Their "relative magnitudes" restriction bounds how much the *change* in the trend violation after treatment can exceed the largest change observed before it, by a factor $\bar{M}$, and the resulting confidence sets account for sampling uncertainty in the pre-period estimates rather than treating them as known.
- Report the *breakdown value*: how large $\bar{M}$ would have to be before your conclusion reverses.
- This turns an untestable assumption into a quantitative claim, exactly the move sensitivity analysis made in Week 6.
- `HonestDiD`.

Honest Inference About Pre-Trends (cont.)

Inference reminder

Bertrand, Duflo, and Mullainathan (2004): cluster on the unit at which treatment varies. With few clusters, use CR2 or a wild cluster bootstrap (Week 2).

A DID Checklist ★

1. State the estimand: ATT for whom, over which post-treatment window?
2. Plot the raw group means over time. Before any regression.
3. Is timing staggered? If so, do not use plain TWFE. Use a modern estimator, and report the TWFE estimate only for comparison.
4. Report an event-study plot with pre-period coefficients, using Sun–Abraham or Callaway–Sant'Anna rather than raw TWFE leads.
5. Justify the scale (levels or logs) on substantive grounds.
6. Cluster on the treatment-assignment unit; report the number of clusters.
7. Report a sensitivity analysis for parallel trends.
8. Placebo: a pre-treatment outcome, or an untreated outcome that the policy should not affect.

Looking Ahead

- Fixed effects difference away time-invariant confounders, a real gain, bought with precision and with a new assumption about trends.
- The staggered-adoption literature is a case study in how a method can be standard for decades and still be wrong. Read papers published before 2021 with this in mind, including your own field's classics.
- **Next week**: what to do when there is only *one* treated unit, and "the control group" has to be constructed.
- Notice that conditional parallel trends brought Week 6 back inside a DID: once trends are only parallel given covariates, you need a weighting or outcome model again, and all of Week 6's cautions return with it.

Problem Set 4

Assigned today, due Monday November 9. Covers Weeks 7 and 9.

References

- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 5]
- Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1): 249–275.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91(6): 3253–3285.
- Callaway, Brantly and Pedro H.C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2): 200–230.
- de Chaisemartin, Clément and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110(9): 2964–2996.
- Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2): 254–277.
- Rambachan, Ashesh and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90(5): 2555–2591.

References (cont.)

- Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends." *American Economic Review: Insights* 4(3): 305–322.
- Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences?" *Journal of Econometrics* 235(2): 2218–2244.
- Sun, Liyang and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225(2): 175–199.